

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed.
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Supplement to: J M Siegel. Sleep and Sleep Disorders 4. Sleep function: an evolutionary perspective. *Lancet Neurol* 2022; **21**: 937–46.

Appendix: REM, non-REM, and total sleep amounts in homeotherms

Mammals are arranged according to ascending REM sleep amounts. Mammals in which both non-REM and REM states have been discriminated are listed.

Birds

23 species¹

Mammals

52 species²⁻⁵

Placentals

47 species

	Total sleep (h)	REM time (h)	Non-REM time (h)
Donkey	3.1	0.4	2.7
Giraffe	4.6	0.4	4.2
Tree hyrax	5.3	0.5	4.8
Rock hyrax	5.4	0.5	4.9
Horse	2.9	0.6	2.3
Sheep	3.8	0.6	3.2
Goat	5.3	0.6	4.7
Gray hyrax	6.3	0.6	5.7
Domestic cattle	4.0	0.7	3.3
Grivet	10.0	0.7	9.3
Guinea pig	9.4	0.8	8.6
Degu	7.7	0.9	6.8
Rabbit	8.4	0.9	7.5
Tapir	4.4	1.0	3.4
Baboon	9.4	1.0	8.4
Galago	9.8	1.1	8.7
Patas monkey	10.9	1.1	9.8
Cotton rat	11.3	1.1	10.2
Macaque	10.1	1.2	8.9
Genet	6.3	1.3	5.0
Lesser short-tailed shew	9.1	1.4	7.7
Squirrel monkey	9.6	1.4	8.2
Chimpanzee	9.7	1.4	8.3
House mouse	12.5	1.4	11.1
Chinchilla	12.5	1.5	11.0
Phalanger	13.7	1.8	11.9
Owl monkey	17.0	1.8	15.2
Human	8.0	1.9	6.1
Mongolian gerbil	14.2	1.9	12.3
Eastern American mole	8.4	2.1	6.3
Star-nosed mole	10.3	2.2	8.1

Three-toed sloth	14.4	2.2	12.2
Greater short-tailed shrew	14.9	2.3	12.6
Tenrec	15.6	2.3	13.3
Pig(wild boar)	9.1	2.4	6.7
Red fox	9.8	2.4	7.4
Mole rat	10.6	2.4	8.2
Norway rat	13.0	2.4	10.6
Mountain beaver	14.4	2.4	12.0
Tree shrew	8.9	2.6	6.3
Desert hedgehog	10.3	2.7	7.6
Dog	10.1	2.9	7.2
Golden hamster	14.3	3.1	11.2
Long-nosed armadillo	17.4	3.1	14.3
cat	12.5	3.2	9.3
Thirteen-lined ground squirrel	13.8	3.4	10.4
European hedgehog	10.1	3.5	6.6

Marsupials

3 species

	Total sleep (h)	REM time (h)	Non- REM time (h)
Phalanger	13.7	1.8	11.9
North American opossum	18	4.9	13.1
Thick-tailed opossum	19.4	6.6	12.8

Monotremes

2 species

	Total sleep (h)	REM time (h)	Non- REM time (h)
Short-beaked echidna	15.0	6.9	8.1
Platypus	14.2	8	6.2

References

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